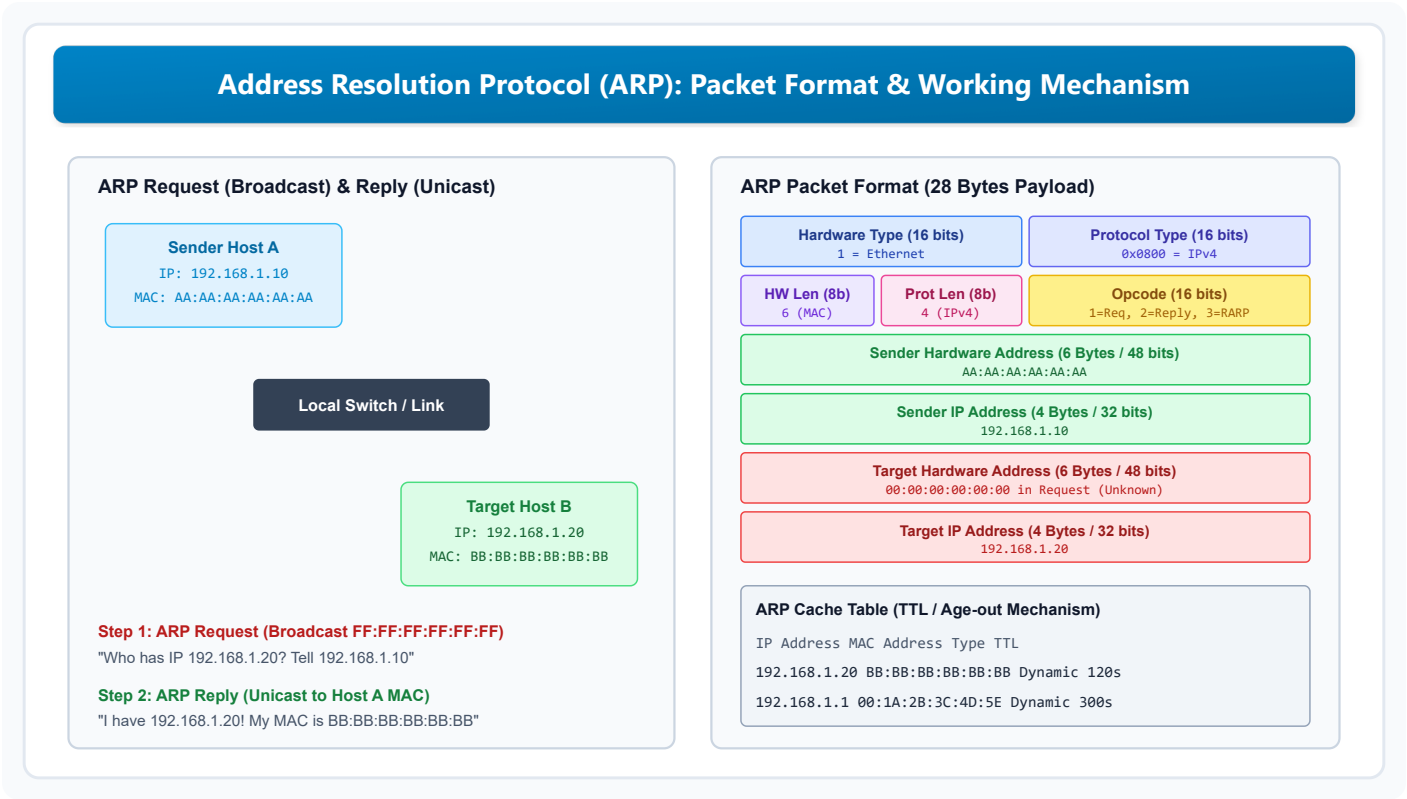


Module 08: Address Resolution Protocol (ARP) and RARP

Unit 3: Network Layer L3 to L2 Address Resolution

Logical IP to Physical MAC Address Binding, ARP Request Broadcast & Reply Unicast, 28-Byte Packet Format, 4 Operational Scenarios, ARP Cache, Gratuitous ARP & RARP.



1. ARP Core Principle

Address Mapping: Known Layer 3 IP Address -> Resolved Layer 2 MAC Address

ARP Request: Broadcast (FF:FF:FF:FF:FF:FF) - Sent to all hosts on local link

ARP Reply: Unicast - Sent strictly to requesting host's known MAC

2. ARP 28-Byte Packet Anatomy

Field	Bit Width	Description / Sample Value
Hardware Type	16 bits	1 for Ethernet link
Protocol Type	16 bits	0x0800 for IPv4 protocol
HW / Prot Length	8 + 8 bits	6 bytes (MAC) / 4 bytes (IPv4)
Opcode	16 bits	1 = Request, 2 = Reply, 3 = RARP Req, 4 = RARP Rep
Sender HW / IP Addr	48 + 32 bits	Sender's MAC (6B) and IP (4B)
Target HW / IP Addr	48 + 32 bits	Target MAC (00:00:... in Request) and Target IP (4B)